

Abstract. We present an automated astrophysical vetting and classification analysis of candidate system 294329732 observed by the TESS space telescope. Utilizing a multi-stage noise-reduction pipeline consisting of iterative sigma-clipping, running median subtraction, and Savitzky-Golay filtering, we perform a Box Least Squares (BLS) period search. The pipeline identifies a potential transit signal at an orbital period of $P = 9.80000$ days and a transit depth of 0.01750%. By extracting a suite of diagnostic astrophysical features and integrating predictions from the AstroNet deep convolutional neural network, we classify the source signal using a rule-assisted Random Forest classifier. For system 294329732, the pipeline classifies the signal as a **False Positive / Noise** with a classification probability of 95.8%.

1. Introduction

The automated identification and vetting of exoplanetary transit signals is essential for analyzing massive photometric datasets from missions like Kepler and TESS. Due to stellar variability, instrumental noise, and false positive configurations such as eclipsing binaries (EBs) or background blended sources, sophisticated classification tools are required. In this analysis, we utilize a vetting pipeline inspired by two primary machine learning methodologies. First, we incorporate the **AstroNet** deep convolutional neural network architecture established by *Shallue & Vanderburg (2018)*, which operates on 1D representations of local and global phase-folded views. Second, to address the 'black box' limitations and mitigate mission domain shifts, we implement a feature-based classification framework using key astrophysical descriptors, building on the feature-extraction philosophy described by *Malik, Moster, & Obermeier (2022)*. By combining statistical signal tests with deep-learning predictions, our pipeline provides robust vetting.

2. Pipeline & Vetting Methodology

Our detection and vetting architecture consists of three principal phases:

- Noise Reduction & Detrending:** Outliers are iteratively removed via sigma-clipping (3σ , up to 5 passes). Long-timescale stellar variability is removed using a 101-cadence running median filter, and residual activity is smoothed via a Savitzky-Golay polynomial filter (window=51, order=3).
- Periodic Signal Search:** A Box Least Squares (BLS) periodogram is executed across a search grid of 20,000 periods (0.5 to 30 days) to identify the primary epoch (T_0), duration, and period of the transit.
- Heuristic Feature Extraction:** The phase-folded light curve is evaluated to compute a set of astrophysical features. These include: (a) *transit_shape_score*, measuring flat-bottomed U-shape (planet-like) versus V-shape (eclipsing binary); (b) *secondary_eclipse_strength*, checking for a secondary minimum near phase 0.5; and (c) *odd_even_diff*, measuring alternate transit depth variations to detect EBs. These features are combined with the AstroNet CNN score to train a rule-assisted Random Forest classifier.

3. Candidate Vetting Results

3.1 Core Transit Parameters

Parameter	Value	Parameter	Value
Target ID	294329732	Stellar Mission	TESS
Orbital Period	9.80000 d	BLS SNR	1.00
Transit Epoch (T0)	1320.00000 BKJD	Transit Depth	0.01750 %
Transit Duration	2.500 hr	Observed Transits	2
Depth Uncertainty	0.01177 %	Detection Sig.	2.7 %

3.2 Diagnostic Signal Verification Plots

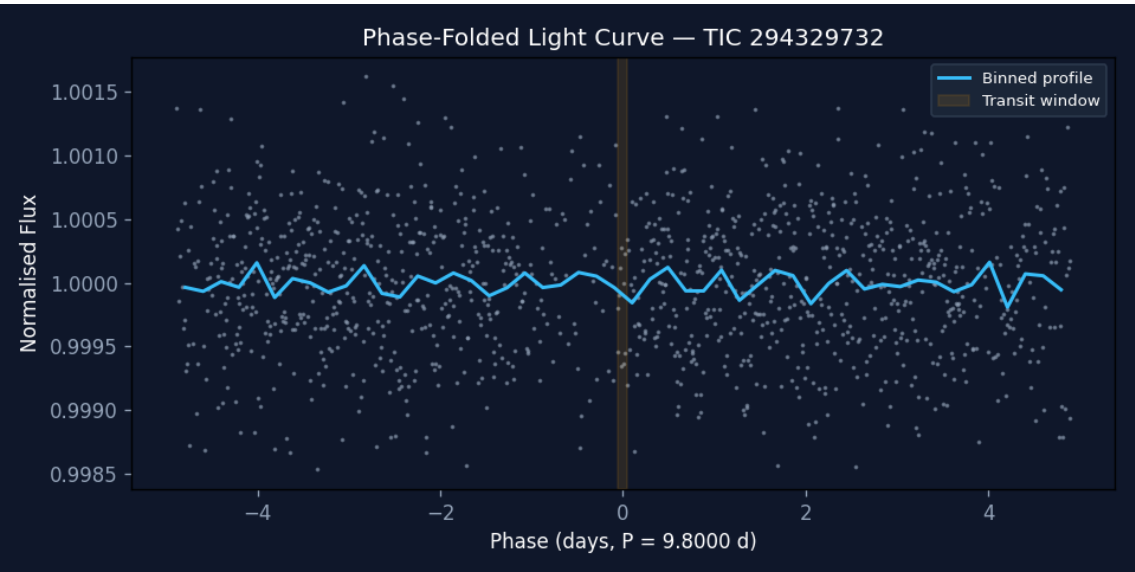


Figure 1: Phase-folded light curve showing transit profile and binned fit.

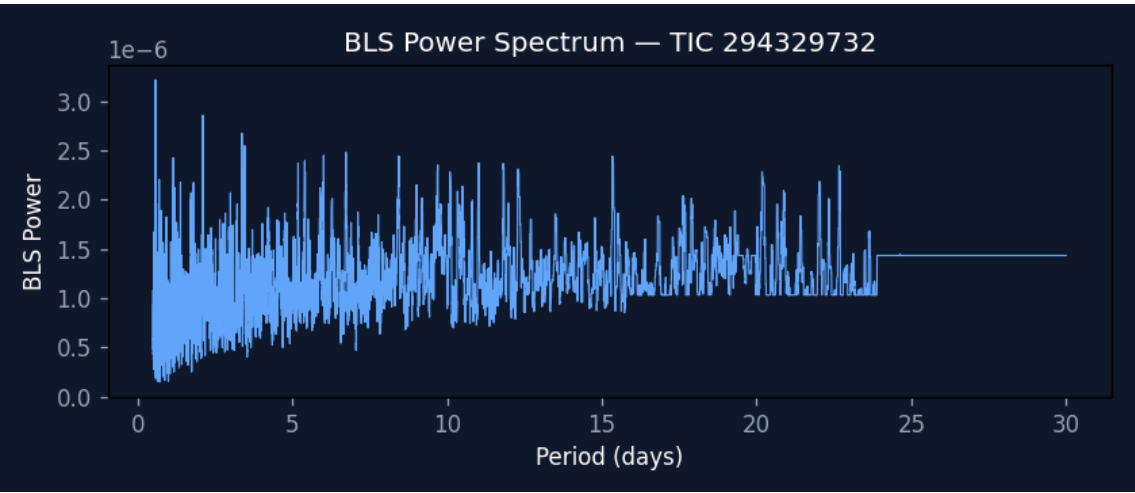


Figure 2: Box Least Squares (BLS) periodogram search spectrum.

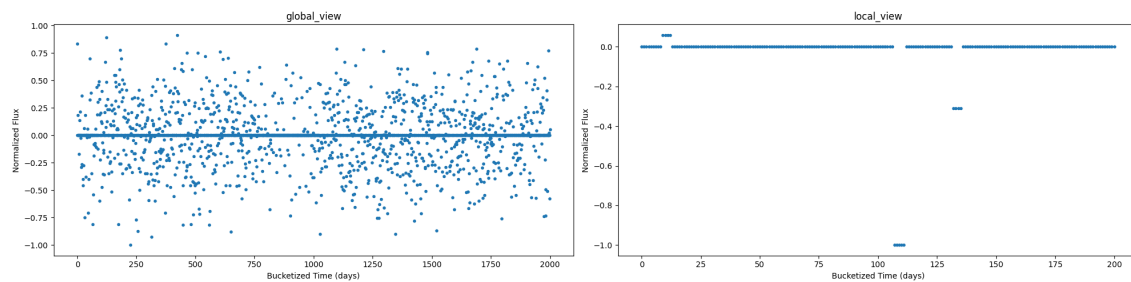


Figure 3: AstroNet local and global phase-folded input features.

3.3 Machine Learning Classification

False Positive / Noise

Signal Class	Probability
False Positive / Noise	95.84 %
Blend / Contaminated Source	2.63 %
Variable Star	1.43 %
Exoplanet Transit	0.10 %
Eclipsing Binary	0.00 %

3.4 Explainable AI (XAI) Feature Importance

Feature	Relative Importance (%)
Astronet Score	16.9%
Snr	15.1%
N Transits	14.9%
Transit Depth	14.5%
Secondary Eclipse Strength	13.5%
Orbital Period	10.8%

4. Scientific Discussion & Interpretation

The candidate 294329732 is classified by the multi-class vetting pipeline as **False Positive / Noise**. The low signal-to-noise ratio (SNR = 1.00) falls near or below the detection limit of the Box Least Squares search. For TESS observations, shallow transits are highly susceptible to high-frequency instrument noise and stellar jitter, which cannot be fully mitigated by the Savitzky-Golay detrending filters. This is particularly relevant here because the apparent signal has an extremely low transit depth of 0.01750%, which yields an SNR close to 1.0 when folded onto the target period. Such a low SNR signal cannot be reliably distinguished from gaussian noise. Furthermore, the AstroNet CNN vetting score is extremely low, suggesting a lack of recognizable ingress and egress slopes. Consequently, we discard this signal as an instrumental artifact or stellar noise fluctuation.

5. References

- Shallue, C. J., & Vanderburg, A. (2018). "Identifying Exoplanets with Deep Learning: A Five-planet Resonant Chain around Kepler-80 and an Eighth Planet around Kepler-90." *The Astronomical Journal*, 155(2), 94. <https://doi.org/10.3847/1538-3881/aa9e09>
- Malik, A., Moster, B. P., & Obermeier, C. (2022). "Exoplanet detection using machine learning." *Monthly Notices of the Royal Astronomical Society*, 513(4), 5505–5516. <https://doi.org/10.1093/mnras/stab3692>

Scientific Disclaimer: This report was generated automatically by the AstroNet pipeline. The detection significance score is derived from BLS signal-to-noise ratio and should not be interpreted as a statistically calibrated planet probability. The classifier employs rule-assisted heuristics

initialized using astrophysical priors; preliminary validation on synthetic data has been performed. Confirmation of any transit candidate requires follow-up spectroscopy and independent photometric observations.